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The Editor-in-Chief
Automation in Construction
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Dear Editor,

We are pleased to submit our manuscript entitled “From Passive Members to Self-Aware Steel Structures: A Digital-Twin and Machine-Learning Framework for Structural Cognition and Autonomous Health Monitoring” for consideration as a research article in *Automation in Construction*.

Conventional structural health monitoring of steel structures excels at acquiring and storing data, but the interpretation of that data remains an offline, expert-driven task. The structure itself senses, yet it does not understand its own condition. Our manuscript addresses this gap by proposing *Structural Cognition*: a paradigm in which a steel structure perceives, processes, interprets and autonomously reports its own mechanical state through a permanent link to a continuously updated digital twin.

The contribution is threefold. First, we formalise Structural Cognition and define a six-level model of structural self-awareness mapped onto the established Rytter damage-identification hierarchy. Second, we propose a five-layer reference architecture that integrates distributed multi-sensor fusion, a finite-element digital twin and a supervised/unsupervised machine-learning layer. Third, we specify a reproducible six-step methodology and an instrumented braced-frame demonstrator for validation.

We believe the work fits the scope of *Automation in Construction*, which covers digital twins, artificial intelligence, sensing and autonomous systems for the monitoring, operation and maintenance of civil and building infrastructure. By moving monitoring from data acquisition toward autonomous interpretation, the framework targets safer, lower-cost and more reliable steel infrastructure.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission. The authors declare no conflict of interest.

Sincerely,

[Corresponding Author Name]
on behalf of all authors